LUKE R. HORTON

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EDUCATION

Aerospace Engineering, B.S. | San Diego State University **Aug. 2022-May 2026**

- **Coursework:** Aerospace Structural Dynamics, Aerospace Structural Mechanics and Materials**,** Spacecraft Attitude Dynamics and Control, Rocket and Space Propulsion, Feedback Flight Control, Aircraft Stability and Control, Astrodynamics, Aircraft Propulsion Systems, Low and High Speed Aerodynamics, Programming and Numerical Methods in Aerospace

EXPERIENCE

Next Generation-Carrier Based Strike Fighter **San Diego State University, Senior Capstone**
Performance Lead & Systems Engineer *Fall 2025- Spring 2026*

- Led performance and test-planning analysis for a conceptual Navy carrier-based strike fighter: translated mission/carrier requirements into verification-style metrics, owned weight estimation, wing sizing, and takeoff/landing analyses, and coordinated with subsystem leads to deliver PDR technical memos.
- Built and validated physics-based sizing/performance models (drag polar, T/W–W/S trades, takeoff/landing constraint sets), ran structured parametric sweeps and sensitivity checks to quantify carrier-suitability margins and drive configuration trade decisions.

Wing in Ground Effect **San Diego State University, Senior Design**
Project Manager ***Spring 2026***

- Designed and test-engineered a GEV prototype through a rapid design–build–test–iterate cycle: built CAD, integrated sensors for data capture, executed controlled near-surface tests, reduced data, and refined the design based on results.
- Led requirements/schedule and anomaly troubleshooting: converted performance goals into airfoil/planform/CG-prop trades, diagnosed test issues, implemented fixes, and validated improvements through re-test and documentation.

SDSU Rocket Project **San Diego State University**
External Structures Engineer *Fall 2023- Fall 2024*

- Performed structural analysis and iterative testing on rocket launch lugs, refining SolidWorks CAD/drawings for manufacturability and subsystem integration while collaborating in cross-team design reviews to meet structural and schedule constraints.

DMG Corporation **San Diego, CA**
Inside Sales Engineer Intern *Sept. 2025- Present*

- Provide operational support to outside sales engineers by generating quotes, securing vendor pricing, processing sales orders and POs, coordinating with manufacturers, and communicating status updates.

San Diego Sales Engineers (National Society of Sales Engineers) **San Diego State University**
President *Fall 2025- Present*

- Planned and organized networking sessions and technical workshops on sales engineering, while also participating in panels and a conference competition to strengthen skills and industry connections.

Participant, National Sales Engineering Conference (NSEC) 2024 & 2025 *Fall 2024 & 2025*

PERSONAL PROJECTS

Welding Project *Fall 2023*

- Co-led the design and fabrication of a custom 800-pound stage for charity events, utilizing SolidWorks and employing advanced welding techniques to ensure structural integrity.

Modular Stage System *Fall 2022*

- Led the design and construction of a 700 ft² modular stage emphasizing repeatable assembly, interchangeable modules, and robust connection.

LEADERSHIP

President, *Kappa Alpha Order* *2024*

- Led a 170+ member chapter, directing an executive board and 20+ chairs while managing a $200K+ budget and chapter operations; earned IFC Most Improved Chapter (2024) for measurable gains in membership, engagement, and overall performance.

SKILLS

CAD & Design: SolidWorks, GD&T, Design for Manufacturability, Rapid Prototyping
Simulation & Analysis: ANSYS, NX Nastran (FEA), MATLAB, Simulink
Programming & Automation: Python, Git/GitHub
Documentation & Tools: LaTeX, Bluebeam Revu, Microsoft Office Suite, Google Suite
Manufacturing & Fabrication: MIG/TIG welding, machining, woodworking, prototype build & assembly